

2 All organisms contain one or more cells.

(a) (i) In which of the following is a cell membrane present?

(1)

- ☐ A animal cells only
- ☐ B animal and plant cells only
- ☐ C plant and prokaryotic cells only
- ☐ D animal, plant and prokaryotic cells

(ii) In which of the following is a cell wall present?

(1)

- ☐ A plant cells only
- ☐ B animal and plant cells only
- ☐ C plant and prokaryotic cells only
- ☐ D animal, plant and prokaryotic cells

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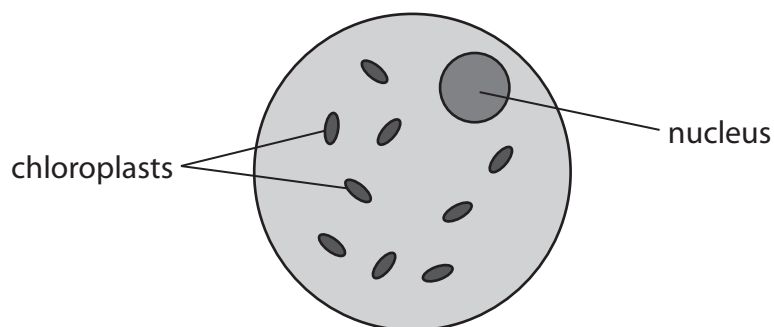
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(b) Sailor's eyeball (*Valonia ventricosa*) is a single-celled, spherical organism.

The diagram shows one of these cells.



(i) The diameter of this cell is $25\text{ }\mu\text{m}$.

Calculate the magnification of the diagram.

(2)

Answer

(ii) This cell contains chloroplasts.

State the function of these chloroplasts.

(1)

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(iii) This cell is not a prokaryotic cell as it contains chloroplasts.

Give **one** other reason why this organism is not a prokaryotic cell.

(1)

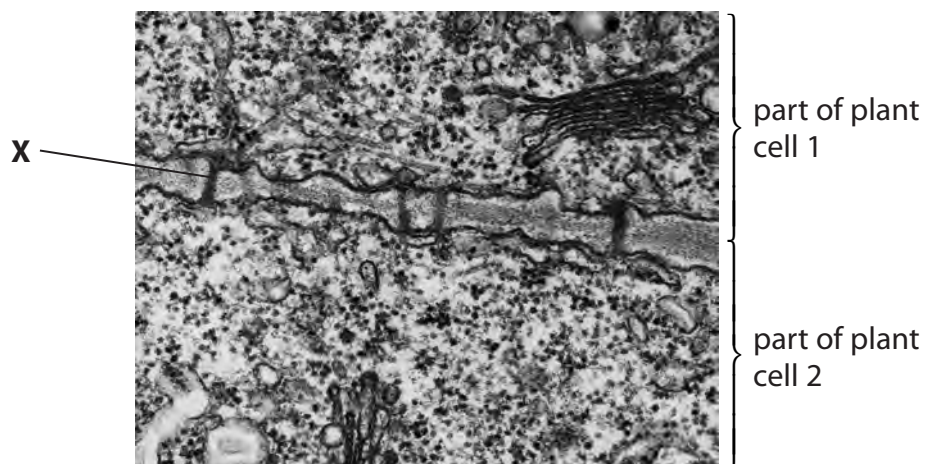
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- (c) The photograph shows part of two adjoining plant cells, as seen using an electron microscope.



(Source: © biophoto associates/Getty Images)

- (i) Name the part labelled **X**.

(1)

- (ii) Explain the function of the part labelled **X**.

(2)

(Total for Question 2 = 9 marks)

